Solution

A1. According to the capacitor energy formula

$$W = \frac{(ne)^2}{2C} \quad \Rightarrow \quad \Delta W = \frac{e^2}{2C} \cdot \left((n+1)^2 - n^2\right).$$

$$\boxed{\Delta W = \frac{e^2}{C}(n + 1/2)}$$

A2. The sum of charges on the capacitor plates is determined by the number n of electrons on the island. The sum of voltages across the capacitors is determined by the voltage source.

By writing the charge conservation law and the sum of voltage drops, we obtain:

$$\begin{cases} q_j - q_g = -ne, \\ V = \dfrac{q_j}{C_j} + \dfrac{q_g}{C_g}. \end{cases}$$

If $C = C_j + C_g$, then after transformations the required expressions are obtained.

$$\boxed{\begin{cases} q_g = \dfrac{C_g}{C} \cdot (VC_j + ne) \\ q_j = \dfrac{C_j}{C} \cdot (VC_g - ne) \end{cases}}$$

A3. The total energy of the system:

$$W_n = \frac{q_g^2}{2C_g} + \frac{q_j^2}{2C_j} = \frac{VC_jC_g + n^2e^2}{2C}.$$

When the number of electrons on the island changes $n \to n + 1$:

$$W_{n+1} = \frac{q_g^2}{2C_g} + \frac{q_j^2}{2C_j} = \frac{VC_jC_g + (n+1)^2e^2}{2C}.$$

The energy increment is $\Delta W = W_{n+1} - W_n = \frac{e^2}{C}(n + 1/2)$.

$$\boxed{\Delta W = \frac{e^2}{C}(n + 1/2)}$$

A4. The work done by the source equals the product of its voltage and the amount of charge that has flowed through it. The amount of charge that has flowed through the source equals the change in the gate charge Δq_g, therefore the source work is:

$$A = V\Delta q_g.$$

According to task A2 $\Delta q_g = eC_g/C$, which gives the answer for the source work.

$$\boxed{A = \frac{C_g}{C}eV}$$

The heat losses are determined by the energy conservation law:

$$Q = A - \Delta W,$$

from which, according to task A2, we obtain the answer.

$$\boxed{Q = \frac{e}{C}\left[VC_g - e(n + 1/2)\right]}$$

A5. The amount of heat released by the resistor equals the product of the resistor voltage and the charge that has passed through it:

$$Q = I_t R_t e$$

Using the expression for Q from point A4, we obtain

$$I_t = \frac{VC_g - e(n + 1/2)}{CR_t}$$

Tunneling is possible when $Q = I_t R_t e > 0$, i.e. when $V > \frac{e}{C_g}(n + 1/2)$, which gives the condition for n.

$$\boxed{I_t = \frac{VC_g - e(n + 1/2)}{CR_t}}$$

$$\boxed{n < \frac{C_g V}{e} - \frac{1}{2}}$$

A6. Differentiating the expression for the gate charge obtained in point A2 with respect to time gives:

$$\dot{q}_g = \frac{C_g}{C}\frac{\Delta n}{\Delta t}e.$$

Taking into account $I = \dot{q}_g$, $I_t = \frac{\Delta n}{\Delta t}e$ we obtain the answer.

$$\boxed{I = \frac{C_g}{C}I_t}$$

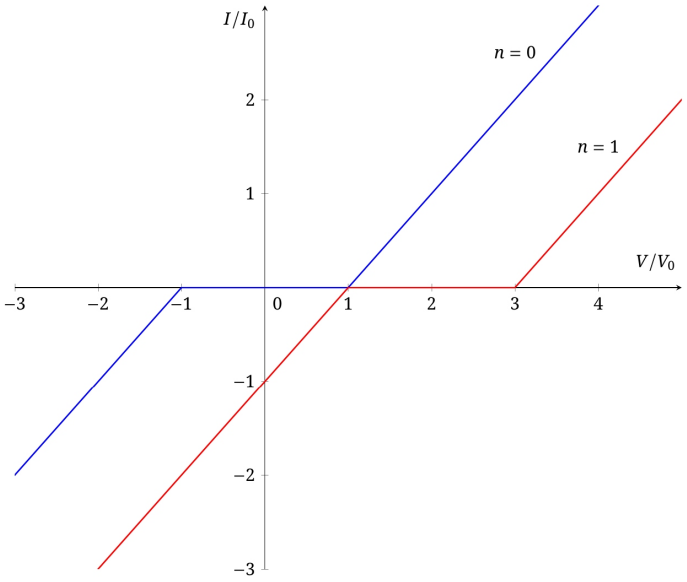
A7. From the two previous tasks:

$$I = \frac{C_g^2}{C^2 R_t}\left[V - \frac{e}{C_g}(n + 1/2)\right] \quad \text{if} \quad V > \frac{e}{C_g}(n + 1/2).$$

Performing similar calculations for the case of decreasing the number n of electrons on the island leads to the possibility of negative current flow:

$$I = \frac{C_g^2}{C^2 R_t}\left[V - \frac{e}{C_g}(n - 1/2)\right] \quad \text{if} \quad V < \frac{e}{C_g}(n - 1/2).$$

Outside the specified intervals, tunneling is impossible, meaning $I(V) = 0$. The required graphs are shown in the figure below.



The current-voltage characteristic of the single-electron box: $V_0 = \frac{e}{2C_g}$, $I_0 = \frac{eC_g}{2C^2R_t}$

B1. The sum of charges on the capacitors equals the charge on the island:

$$C_1(\varphi - V_1) + C_2(\varphi - V_2) + C_g(\varphi - V_g) = -ne.$$

Substituting the expressions for V_1, V_2, we obtain the answer.

$$\boxed{\varphi = \frac{(C_1 - C_2)V/2 + C_gV_g - ne}{C_1 + C_2 + C_g}}$$

B2. First, we express the changes in charges on the capacitor plates when the number of electrons on the island changes:

$$q_1 = C_1(\varphi - V_1) = \frac{C_1}{C}\left[(C_1 - C)V_1 + C_2V_2 + C_gV_g - ne\right]$$

$$\Delta q_1 = -\frac{eC_1}{C}\Delta n.$$

Here and hereafter in part B $C = C_1 + C_2 + C_g$. Similarly

$$\Delta q_2 = -\frac{eC_2}{C}\Delta n, \qquad \Delta q_g = -\frac{eC_g}{C}\Delta n.$$

The work done by the sources is determined by the charge flowing through them. For the branch where tunneling occurs, one must account for the transfer of the tunneling charge itself $q_t = -\Delta ne$. Thus, the total work done by the sources in our case takes the form

$$A = -V_1\Delta q_1 - V_2\Delta q_2 - V_g\Delta q_g + V_1q_t.$$

Substituting the expressions for the charges, we obtain:

$$A = \frac{e\Delta n}{C}\left[V_1(C_1 - C) + V_2C_2 + V_gC_g\right].$$

Taking into account $\Delta n = 1$ and the expressions for V_1, V_2, we write the result.

$$\boxed{A = \frac{e}{C}\left[V_gC_g - V/2\,(C_g + 2C_2)\right]}$$

B3. The total energy of the system

$$W_n = \frac{C_1}{2}(\varphi - V_1)^2 + \frac{C_2}{2}(\varphi - V_2)^2 + \frac{C_g}{2}(\varphi - V_g)^2$$

after substituting the expression for the island potential φ and simplification takes the form

$$W_n = \frac{(ne)^2}{2C} + W_0,$$

where W_0 is a term independent of the number n of electrons on the island.

Then the energy increment resulting from tunneling of one electron

$$\Delta W_n = \frac{e^2}{2C}((\Delta n)^2 + 2n\Delta n) = \frac{e^2}{C}\left(\frac{1}{2} \pm n\right),$$

where the $+$ sign corresponds to an increase in the number of electrons on the island.

The amount of heat released is determined from the energy conservation law:

$$Q = A - \Delta W_n = \pm\frac{e}{C}\left[V_1(C_1 - C) + V_2C_2 + V_gC_g - e(n \pm 1/2)\right].$$

Considering the case of increasing the number of electrons, we obtain the answer.

$$\boxed{Q = \frac{e}{C}\left[V_gC_g - V/2\,(C_g + 2C_2) - e(n + 1/2)\right]}$$

B4. Taking into account the expressions for V_1, V_2, we obtain the following for the heat released in the circuit:

$$Q = \pm\frac{e}{C}\left[V_gC_g - V/2\,(C_g + 2C_2) - e(n \pm 1/2)\right],$$

where the $+$ sign still corresponds to an increase in the number of electrons on the island.

From the inequality $Q > 0$ for the upper signs we obtain:

$$\boxed{V < \frac{V_gC_g - e(n + 1/2)}{C_2 + C_g/2}}$$

B5. For electron tunneling from the island through the first junction we obtain:

$$V > \frac{V_gC_g - e(n - 1/2)}{C_2 + C_g/2}.$$

In the case of tunneling through the second junction, we replace $V \to -V, C_1 \to C_2$ in the expression for the released heat and obtain the condition:

$$Q = \pm\frac{e}{C}\left[V_gC_g + V/2\,(C_g + 2C_2) - e(n \pm 1/2)\right] > 0.$$

From this, for tunneling onto the island we obtain the inequality

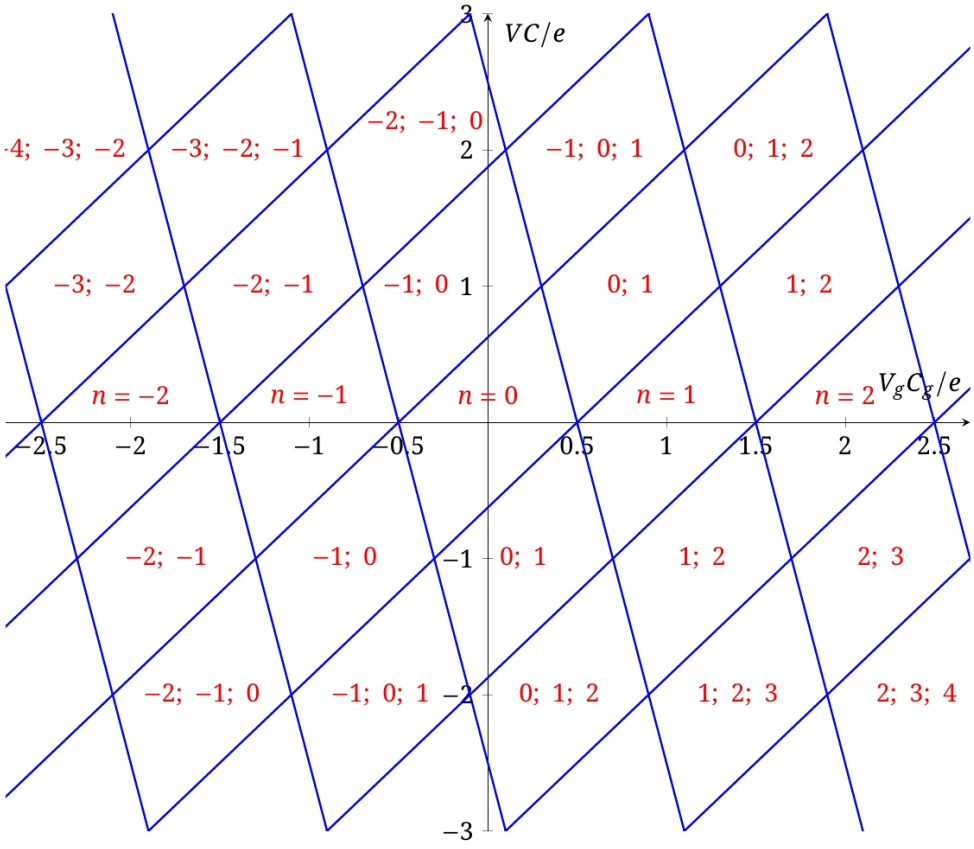
$$V > -\frac{V_gC_g - e(n + 1/2)}{C_1 + C_g/2}.$$

For tunneling from the island:

$$V < -\frac{V_gC_g - e(n - 1/2)}{C_1 + C_g/2}.$$

Thus, in the coordinates V, V_g the regions of possible tunneling are separated by two families of straight lines: with slope coefficients $k_1 = \frac{C_g}{C_2+C_g/2}$ and $k_2 = -\frac{C_g}{C_1+C_g/2}$.

Each region forms a parallelogram. The parallelograms containing the abscissa axis correspond to a single equilibrium number n. These regions represent Coulomb blockade — the impossibility of current flow through the transistor. In other regions, the resistance becomes finite, and current can flow.



The V, V_g voltage diagram of the transistor

B6. Positive current corresponds to tunneling from the second bank to the island, and then from the island to the first bank. Let us consider both these processes and use the equality $Q_\alpha = I_\alpha R_t e;\ \alpha = 1, 2$ — the tunneling process number:

$$\begin{cases} I_1 R_t e = \dfrac{e}{C}\left[V(C_1 + C_g/2) + V_g C_g - e(n + 1/2)\right], \\ I_2 R_t e = -\dfrac{e}{C}\left[-V(C_2 + C_g/2) + V_g C_g - e(n + 1 - 1/2)\right]. \end{cases}$$

For the total current flowing through the transistor, determined by the total tunneling time across both junctions, we obtain $I = (I_1^{-1} + I_2^{-1})^{-1}$. Introducing the quantity $Q_g = V_g C_g - e\bar{n}$, where $\bar{n} = (n + (n+1))/2$ is the average number of electrons on the island, then after simplification the dependence takes the form:

$$I(V, Q_g) = \frac{C^2V^2/4 - Q_g^2}{VC^2R_t}.$$

Particularly for $n = 0, \bar{n} = 1/2$, we obtain the required dependence.

$$\boxed{I(V, V_g) = \frac{1}{4R_t}\left[V - \frac{4}{VC^2}\left(C_g V_g - e/2\right)^2\right]}$$

C1. The graph clearly shows the "Coulomb diamonds" obtained in the task B5. The measurement of period ΔV_g allows determination of the gate capacitance, while the maximum voltage $V_{\max}$ gives the total capacitance. From the graph $\Delta V_g = 7$ mV, $V_{\max} = 0.4$ mV. Hence $C_g = e/\Delta V_g = 23$ aF, $C = e/V_{\max} = 400$ aF, $C_1 = C_2 = (C - C_g)/2 = 190$ aF.

$$\boxed{C_1 = C_2 = 190 \text{ aF}}$$

$$\boxed{C_g = 23 \text{ aF}}$$

C2. The change in probability of having n electrons on the island is related to possible processes $n+1 \to n$ and $n-1 \to n$ that increase this probability, as well as reverse processes that decrease it:

$$\boxed{\mathrm{d}p(n)/\mathrm{d}t = p(n-1)\cdot\Gamma_{n-1\to n} + p(n+1)\cdot\Gamma_{n+1\to n} - p(n)\cdot(\Gamma_{n\to n-1} + \Gamma_{n\to n+1})}$$

C3. From the previous point it is known that for arbitrary n the following holds:

$$p(n-1)\cdot\Gamma_{n-1\to n} - p(n)\cdot\Gamma_{n\to n-1} = p(n)\cdot\Gamma_{n\to n+1} - p(n+1)\cdot\Gamma_{n+1\to n}.$$

Taking into account the definition of n_0:

$$p(n_0-1)\cdot\Gamma_{n_0-1\to n_0} = p(n_0)\cdot\Gamma_{n_0\to n_0-1}.$$

From this it follows that for all n less than n_0, the following holds:

$$p(n-1)\cdot\Gamma_{n-1\to n} = p(n)\cdot\Gamma_{n\to n-1}.$$

Indeed, relations of this type imply vanishing of the left-hand side of the original equality for arbitrary n. Consequently, the right-hand side, corresponding to n differing by unity, also vanishes. This leads to the answer.

$$\boxed{A = \frac{\Gamma_{n\to n+1}}{\Gamma_{n+1\to n}}}$$

C4. The expressions for the released heat were obtained in part B and take the form:

$$Q^{(i)}_{n\to n\pm1} = \pm\frac{e}{C}\left[(-1)^i\, VC/2 + (V_g C_g - e/2) - e(n - 1/2 \pm 1/2)\right].$$

Considering $|V| \ll e/C, |V_g - e/2C_g| \ll e/C_g$, for both tunneling junctions, increasing the number n of electrons when $n > 0$ leads to a negative value of the released heat $|Q_{n\to n\pm1}| > e^2/C \gg kT$. The same inequality occurs when decreasing n for $n < 1$.

Thus, under the specified conditions, a large positive value appears in the exponent, and the unity in the denominator can be neglected. This leads to the required result.

C5. Using the result from point C3:

$$p(0)\cdot\Gamma_{0\to1} = p(1)\cdot\Gamma_{1\to0}$$

Taking into account the normalization condition $p(0) + p(1) = 1$, we express the probabilities $p(0)$ and $p(1)$.

$$\boxed{\begin{cases} p(0) = \dfrac{\Gamma_{1\to0}}{\Gamma_{0\to1} + \Gamma_{1\to0}} \\ p(1) = \dfrac{\Gamma_{0\to1}}{\Gamma_{0\to1} + \Gamma_{1\to0}} \end{cases}}$$

C6. The current through the transistor is determined by the number of tunnelings through one of the junctions (tunnelings through the other occur such that the number of electrons on the island remains unchanged):

$$I = e\left(p(1)\cdot\Gamma^{(1)}_{1\to0} - p(0)\cdot\Gamma^{(1)}_{0\to1}\right) = -e\left(p(1)\cdot\Gamma^{(2)}_{1\to0} - p(0)\cdot\Gamma^{(2)}_{0\to1}\right).$$

Note that the transition from one expression to another is easily accomplished using

$$p(0)\cdot\Gamma_{0\to1} = p(1)\cdot\Gamma_{1\to0}.$$

Substitution of the expressions for probabilities $p(0), p(1)$ leads to the required result.

$$\boxed{I = e\frac{\Gamma^{(2)}_{0\to1}\Gamma^{(1)}_{1\to0} - \Gamma^{(1)}_{0\to1}\Gamma^{(2)}_{1\to0}}{\Gamma^{(1)}_{0\to1} + \Gamma^{(2)}_{0\to1} + \Gamma^{(1)}_{1\to0} + \Gamma^{(2)}_{1\to0}}}$$